

DroneVibeAnalyzer Engineering Report

Health Dashboard

Overall Score : 76/100
Status : WARNING
Vibration Score : 90/100
Motor Score : 90/100
FFT Score : 50/100
FTN Score : 60/100

Flight Assessment

Issue : Resonance peak detected
Issue : Weak notch tracking
Recommendation : Review FFT analysis
Recommendation : Verify harmonic notch settings
Recommendation : Verify dynamic notch configuration

Vibration Analysis

Mean X : 7.95
Mean Y : 5.25
Mean Z : 5.43
Max X : 39.34
Max Y : 25.05
Max Z : 29.13

FFT Peak Detection

Peak 1 : 20.49 Hz | Amplitude : 60.09
Peak 2 : 46.68 Hz | Amplitude : 46.32
Peak 3 : 61.08 Hz | Amplitude : 39.61
Peak 4 : 72.26 Hz | Amplitude : 33.59
Peak 5 : 34.72 Hz | Amplitude : 33.51

FTN Statistics

Mean NF1 : 0 Hz
Min NF1 : 0 Hz
Max NF1 : 0 Hz

IMU Information

samples : 26511
gx_mean : 0.0
gy_mean : -0.0
gz_mean : -0.01

RCOU Information

samples : 3535
min : 666.67
max : 1049.17
mean : 869.36
span : 382.5
frame_type : QUAD
motor_count : 4
m1_max : 1687
m2_max : 1709
m3_max : 1616
m4_max : 1603
m5_max : 0
m6_max : 0
m1_avg : 1322.87
m2_avg : 1306.26
m3_avg : 1300.9
m4_avg : 1286.11
m5_avg : 0.0
m6_avg : 0.0
m1_std : 271.83
m2_std : 257.57
m3_std : 253.52
m4_std : 239.21
m5_std : 0.0
m6_std : 0.0